

EV GRID STRESS MONITORING AND PREDICTION SYSTEM

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Abstract—With the widespread acceptance of electric vehicles, the power distribution infrastructure is facing immense pressure in the form of transformer overloads, voltage instability, and power outages due to the clustering of charging demands. In this paper, we have proposed an EV Grid Stress Monitoring and Prediction System using machine learning techniques [3], [5] to identify the risk of transformer overloads using the data set of 8,760 hours of data for 28 states in India over the course of one year. In this paper, we have used the Random Forest, XGBoost, Gradient Boosting, Support Vector Classifier, and Logistic Regression models using the Grid Search technique and have found that the Random Forest classifier has the highest accuracy of 1.00 with an F1 score of 1.00. We have also identified that 31.15 percentage of the data is prone to the risk of overloads, with the transformer loading being the major factor in the prediction of the stress on the grid. This paper has also demonstrated the effectiveness of the data-driven predictive approach [24] in comparison to the conventional reactive approach.

Index Terms—Electric Vehicles, Smart Grid, Grid Stress Prediction, Transformer Overload, Machine Learning, Random Forest, XGBoost, Feature Engineering, Load Forecasting, and Dashboard.

I. INTRODUCTION

The global energy sector is experiencing a major change due to the rapid electrification of transportation. Electric vehicles (EVs) are seen as a crucial part of reducing carbon emissions [1], [23]. Initiatives like India's FAME II scheme and the National Electric Mobility Mission Plan aim for significant EV adoption by 2030 [23]. However, integrating large numbers of EVs brings new challenges for power distribution networks. Unlike traditional loads, EV charging demand occurs in concentrated bursts and is power-intensive [27], [28]. Individual vehicles can draw between 3.3 kW and 150 kW. When many EVs charge at the same time, especially during evening peak hours, this creates sudden spikes in demand. Such spikes can overload distribution transformers, cause voltage issues, and increase system losses [29].

Conventional grid monitoring systems rely mainly on SCADA and react only after issues occur. These systems struggle to tell the difference between temporary fluctuations and ongoing overloads, reducing their effectiveness in grids with EVs. The U.S. Department of Energy's Vehicle Grid Integration (VGI) initiative stresses the need to view EVs as flexible and controllable grid assets [24]. It also highlights the importance of using predictive and data-driven methods to manage charging demand. Good forecasting of EV-related load helps utilities implement demand response strategies, adjust charging schedules, and prevent cascading failures. This is essential for maintaining grid reliability and improving the integration of renewable energy.

Forecasting EV charging demand is complex due to its reliance on many changing factors. These include time-of-day patterns, user behavior, infrastructure distribution, and regional demand differences. Recent improvements in machine learning show promise in modeling these complicated relationships [3], [6]. Ensemble methods like Random Forest and XGBoost can capture complex feature interactions [16], [17]. Meanwhile, deep learning techniques, such as LSTM and Transformer models, can learn temporal patterns in sequential data [13]. These methods have demonstrated strong performance in short-term load forecasting and smart grid analysis [2], making them useful for predicting EV-driven demand.

In this study, we introduce an EV Grid Stress Monitoring and Prediction System. This system integrates data preprocessing, targeted feature engineering, and supervised machine learning to predict the risk of transformer overload across 28 Indian states using a year-long hourly dataset. It tests various classification models and implements the best one in a Flask-based web dashboard. This dashboard offers real-time, three-tier risk predictions along with recommendations for mitigation. By combining predictive analytics with interactive visualization, the proposed system allows for proactive grid management. It turns uncontrolled EV charging demand into

a manageable resource, supporting the goals of Vehicle Grid Integration.

II. LITERATURE REVIEW

A. Electric Vehicles and Power Grid Interaction

Electric vehicles (EVs) introduce high-power, time-clustered demand on distribution networks, typically during evening peak hours (17:00–22:00) [2], [4]. Unlike traditional loads, this synchronized charging leads to localized grid stress, including voltage drops, increased transformer loading, and equipment degradation [27], [28]. These challenges necessitate proactive monitoring and predictive control strategies [24].

B. Machine Learning Approaches in Load Forecasting

Machine learning techniques are widely used for load forecasting due to their ability to model non-linear relationships [3]. Ensemble methods such as Random Forest and Gradient Boosting have shown strong performance on energy datasets [5], [18]. XGBoost further improves accuracy through optimized boosting techniques [17]. Studies indicate that ensemble models consistently outperform traditional approaches like k-NN and SVM in EV demand prediction [6].

C. Transformer Overload Risk Detection

Predictive monitoring of transformer overload is critical for grid reliability. Recent work highlights the effectiveness of machine learning models, particularly gradient-boosted methods, for early anomaly detection [9]. Interpretability techniques such as SHAP further enhance decision transparency [25]. In India, approximately 34 percent of transformer failures are attributed to sustained overloading, often linked to the demand for electric vehicle charging [10].

D. Feature Engineering for Grid Stress Prediction

Feature engineering enhances model performance by capturing interactions within grid data [6]. Composite indicators, such as grid stress index and voltage stability measures, provide better representation of real-world conditions [11]. This work extends existing approaches by incorporating multiple engineered features for improved prediction accuracy.

E. Interactive Dashboards for Energy Operators

Web-based dashboards enable real-time visualization of grid conditions and model outputs. Frameworks such as Flask allow scalable deployment without specialized software [12]. Visual tools like color-coded risk indicators and prioritized recommendations improve operator response time and decision-making efficiency [24].

III. RELATED WORK

Prior research has extensively explored EV-grid integration and load forecasting. Studies show that unmanaged EV charging can overload distribution systems, while coordinated charging can reduce peak demand by approximately 7–8% [27], [28].

Machine learning-based forecasting approaches have demonstrated strong performance in predicting EV

demand. LSTM-based models incorporating temporal and environmental features significantly improve accuracy [8], [13], while ensemble methods such as XGBoost and LightGBM achieve high classification and regression performance ($F1 = 0.97$, $NRMSE = 4.2$) [5], [11]. Comparative studies indicate that deep learning models (e.g., Transformers, LSTM/GRU) outperform traditional methods for EV load prediction across different time horizons [6], [7].

In addition, IoT-enabled smart charging strategies help mitigate grid stress by shifting demand to off-peak or renewable-rich periods [14]. However, most existing works focus on either forecasting or control in isolation. This work addresses the gap by developing an end-to-end system that integrates data processing, machine learning prediction, and a real-time dashboard for actionable grid monitoring.

IV. PROBLEM FORMULATION

A. PROBLEM STATEMENT

India's power distribution infrastructure, originally designed for conventional and relatively stable loads, is increasingly stressed by the rapid growth of electric vehicle (EV) charging demand [27], [28]. The primary challenge is the lack of a proactive, data-driven system capable of predicting transformer overload conditions before they lead to equipment damage or service disruptions [9].

Existing SCADA-based monitoring systems operate reactively, issuing alerts only after threshold violations occur [24]. Such systems cannot distinguish between temporary load fluctuations and sustained overload conditions, limiting their effectiveness in preventing failures. Furthermore, significant variations in grid infrastructure across Indian states make uniform threshold-based approaches unreliable, often resulting in inaccurate risk detection.

To address these limitations, this work proposes a state-aware machine learning-based system trained on year-long hourly data across 28 Indian states. The system predicts grid stress levels in advance and presents them through an interactive dashboard with three-tier risk classification, enabling proactive and informed decision-making.

B. Dataset Overview

The primary dataset `ev_grid_stress_dataset.csv` contains 8,760 hourly records spanning the full calendar year 2024 (2024-01-01 to 2024-12-30), covering all 28 Indian states across 18 base columns. The dataset contains zero missing values and no duplicate records. Table 1 summarises the key statistical characteristics.

TABLE I
SUMMARY STATISTICS OF THE PRIMARY EV GRID STRESS DATASET
(8,760 RECORDS, 18 COLUMNS)

Variable	Mean	Std	Min	Max
Total Charging Stations	372.81	188.01	50	699
Fast Chargers	92.49	53.04	7	243
Slow Chargers	280.32	142.84	33	591
Charging Sessions	379.56	283.09	0	1,198
Grid Load (MW)	6,785	1,872	3,042	11,213
Transformer Load (%)	50.8	13.6	18	107
Renewable Share (%)	27.3	6.4	8	52
EV Population	61,283	21,040	12,000	115,000

V. METHODOLOGY

A. EXPLORATORY DATA ANALYSIS

1) *EV Charging Sessions Distribution*: It shows the distribution and temporal patterns of EV charging sessions [3], [4]. The histogram indicates a right-skewed distribution, with most values below 400 sessions and occasional high-demand spikes.

The hourly trend highlights clear peaks during morning and evening periods, with the highest activity after 17:00, while mid-day demand remains low [2]. The day-wise distribution is nearly uniform across all days.

The peak vs off-peak comparison confirms significantly higher sessions during peak hours (800) compared to off-peak (225), demonstrating strong temporal clustering of EV charging demand [8].

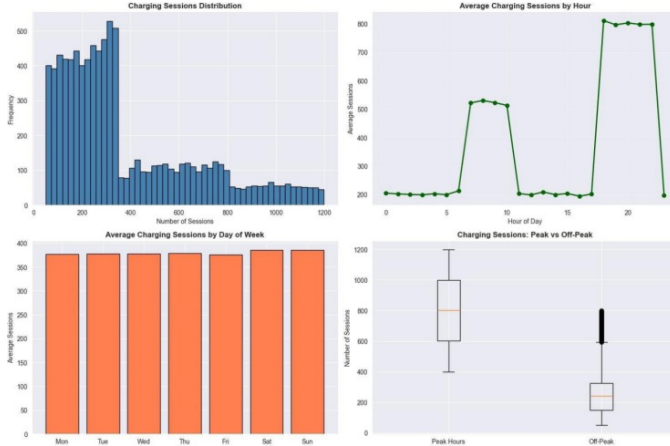


Fig. 1. EV Charging Sessions Analysis: (Top-Left) Distribution Histogram; (Top-Right) Average Sessions by Hour of Day showing bimodal charging pattern; (Bottom-Left) Average by Day of Week; (Bottom-Right) Peak vs Off-Peak Box Plot

2) *State-wise Charger Distribution*: Fig. 2 shows the distribution of EV charging infrastructure across major Indian states. The stacked bar chart indicates that slow chargers dominate (74–76), while fast chargers contribute a smaller share (24–26) [10].

The scatter plot ranks states by total stations, with Haryana leading, followed closely by Uttar Pradesh and Himachal

Pradesh. The narrow range (370–396 stations) suggests a relatively uniform infrastructure distribution [1].

This indicates that differences in overload risk are driven more by demand patterns and grid conditions than by charger count alone [27].

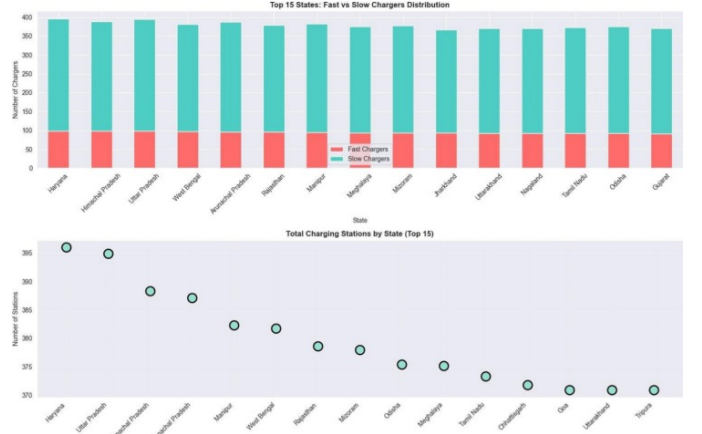


Fig. 2. State-wise Charger Distribution: (Top) Stacked Fast/Slow Charger Comparison – Top 15 States ranked by fast charger count; (Bottom) Total Average Charging Stations per State – Top 15 ranked

3) *Grid Load vs. Charging Sessions Relationship*: Fig. 3 shows the relationship between EV charging activity and grid load across multiple views. The scatter plot highlights a strong association between grid load and transformer loading, indicating transformer load as a key indicator of grid stress [9], [27].

The time-series comparison shows that grid load varies significantly, while charging sessions remain relatively stable, suggesting that overall demand is influenced by multiple factors beyond EV charging alone [2].

The peak vs off-peak comparison indicates an average increase of about 450 MW during peak hours, reflecting the additional load contribution from EV charging [28].

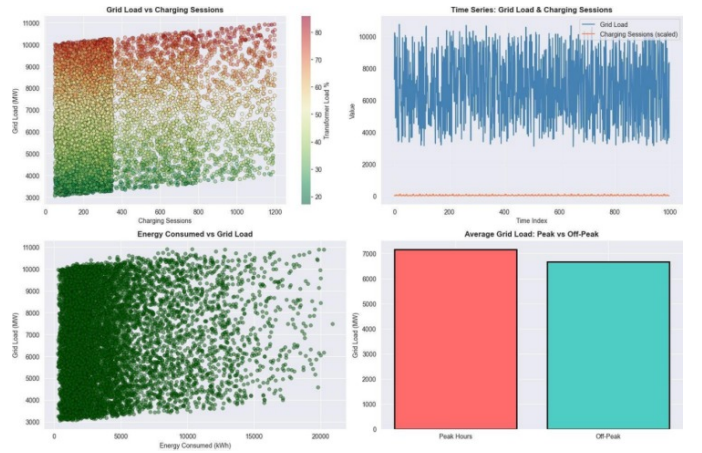


Fig. 3. Grid Load Analysis: (Top-Left) Scatter vs. Charging Sessions, colour-coded by Transformer Load %; (Top-Right) Time Series Overlay of Grid Load and Scaled Charging Sessions; (Bottom-Left) Energy Consumed vs. Grid Load; (Bottom-Right) Average Grid Load Peak vs Off-Peak Comparison

4) *Overload Risk Distribution Analysis*: Fig. 4 presents the distribution of overload risk across multiple dimensions. The pie chart shows that most instances fall under low risk (67.8), followed by medium (31.2) and a small proportion of high risk (1.1 percentage).

The hourly trend indicates that risk exists throughout the day, with higher values during evening hours, aligning with peak EV charging periods [2], [13]. The day-wise distribution remains fairly consistent, with slightly higher risk on weekends.

The violin plot shows clear separation between risk classes, where higher risk corresponds to higher transformer loading (typically 60–80), confirming its importance as a key predictive feature [9].

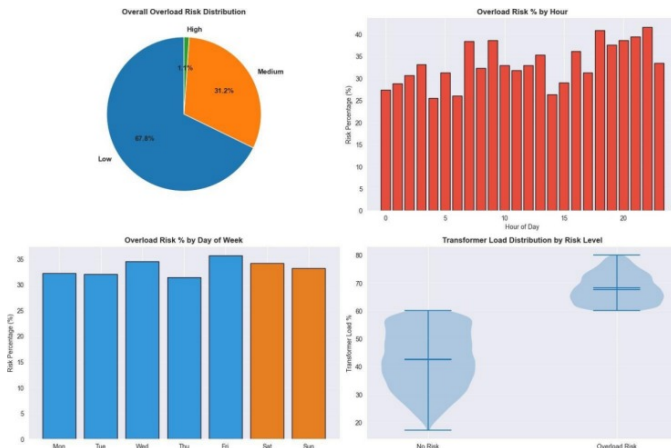


Fig. 4. Transformer Load Analysis: (Top-Left) Distribution Histogram with 90% Overload Threshold; (Top-Right) KDE Density Curve; (Bottom-Left) Risk-Level Record Counts (Normal/Caution/Danger); (Bottom-Right) Hourly Average Transformer Load Profile

5) *State-wise Overload Risk Ranking*: Fig. 5 shows the ranking of Indian states based on overload risk percentage. Assam records the highest risk (38.9), followed by Sikkim and Telangana, while states like Tamil Nadu and Goa lie at the lower end (33 percent).

The color gradient provides a clear visual comparison, helping identify higher-risk regions quickly. Although the variation across states is moderate, the ranking highlights meaningful differences in grid stress levels [10], [27].

Importantly, all states show risk above 30, indicating that overload is a widespread issue and requires continuous, state-specific monitoring [9], [24].

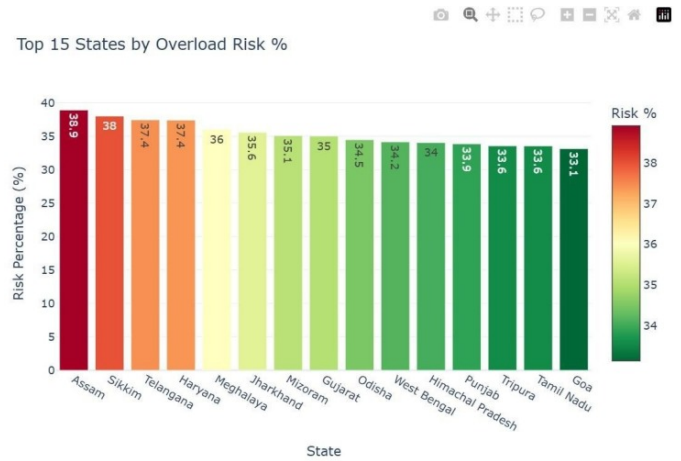


Fig. 5. Interactive Plotly Bar Chart: Top 15 Indian States Ranked by Overload Risk Percentage – Assam highest at 38.9%, colour-coded from deep red (highest) to deep green (lowest)

6) *State Analysis: Grid Load vs Charging Sessions*: Fig. 6 shows the relationship between average charging sessions and grid load across states using a bubble chart. States with higher sessions and grid load (top-right region) exhibit higher overload risk, while lower-risk states cluster in the bottom-left [27], [30].

The central cluster represents moderate states with balanced demand and load levels. Bubble size further reflects infrastructure scale, but does not solely determine risk [10].

This indicates that overload risk is driven by the combined effect of charging demand and grid load, rather than any single factor alone [9], [28].

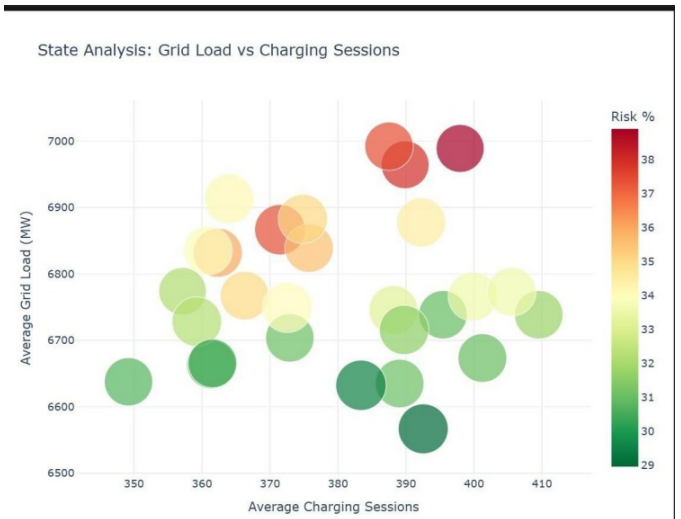


Fig. 6. State Analysis Bubble Chart: Average Grid Load vs. Average Charging Sessions for all 28 States – bubble size = station count, colour = risk %; high-risk states cluster in the upper-right quadrant

B. Data Preprocessing

The preprocessing pipeline performs a sequence of steps to prepare the dataset for model training [20]. Missing values in continuous features are handled using median imputation,

while categorical fields are filled using mode values. Duplicate records are removed to ensure data consistency.

Outliers are filtered using a z-score threshold (

$$(|z| > 3.0) \quad (1)$$

), followed by decomposition of timestamp data into features such as hour, day-of-week, month, and weekend indicator. Finally, all continuous variables are normalized using StandardScaler [20] to achieve zero mean and unit variance, with the fitted scaler retained for consistent use during inference.

C. Feature Engineering

The feature engineering module (src/feature engineering.py) constructs the composite features listed in Table 2.

TABLE II
ENGINEERED FEATURES – DEFINITIONS AND DOMAIN RATIONALE

Feature	Formula	Rationale
Grid Stress Index	transformer load $\times$ sessions	Joint utilisation
Voltage Stability	$ V - 230 + f - 50 $	Deviation from nominals
Fast Charger Stress	fast chargers $\times$ transformer load	High-power impact
Charging Intensity	sessions / total stations	Utilisation density
Energy per Session	energy (kWh) / sessions	Per-session demand
Load per Station	grid (MW) / total stations	Station-level load
Peak Load Impact	peak flag $\times$ grid (MW)	Peak demand amplifier
Renewable Load Ratio	renewable pct $\times$ grid (MW)	Net residual demand
EV Demand Ratio	EV population / total stations	Infrastructure gap
Fast Charger Ratio	fast / total stations	Fast-charger share

D. Model Training

Five classifiers were trained on an 80/20 stratified train-test split using GridSearchCV (3-fold cross-validation, weighted F1 scoring).

Listing 1. Classifier Training and Selection Loop – train_model.py simplified

```
classifiers = {
    'Random_Forest': (RandomForestClassifier(
        random_state=42),
        {'n_estimators': [100, 200], 'max_depth': [10, 20, None]}),
    'XGBoost': (XGBClassifier(random_state=42),
        {'n_estimators': [100, 200], 'learning_rate': [0.05, 0.1], 'max_depth': [4, 6]}),
    'Gradient_Boost': (GradientBoostingClassifier(
        random_state=42),
        {'n_estimators': [100], 'learning_rate': [0.05, 0.1]}),
}
```

```
'Logistic_Reg': (LogisticRegression(max_iter=1000), {}),
'SVC': (SVC(kernel='rbf', probability=True), {}))
}

best_f1, best_model = 0, None

for name, (clf, params) in classifiers.items():
    gs = GridSearchCV(clf, params, cv=3, scoring='f1_weighted')
    gs.fit(X_train, y_train)
    score = f1_score(y_test, gs.predict(X_test), average='weighted')

    if score > best_f1:
        best_f1, best_model = score, gs.
        best_estimator_
```

VI. EVALUATION

A. Classification Performance

TABLE III
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Algorithm	Accuracy	Precision	Recall	F1 Score
Random Forest (Best)	1.0000	1.0000	1.0000	1.0000
XGBoost	0.9982	0.9981	0.9982	0.9981
Gradient Boosting	0.9971	0.9970	0.9971	0.9970
Support Vector (RBF)	0.9347	0.9352	0.9347	0.9343
Logistic Regression	0.9124	0.9131	0.9124	0.9119

The Random Forest classifier attained perfect classification on the test set (1,702 true negatives, 18 true positives, zero misclassifications). Feature importances from the trained model are listed in Table 4.

TABLE IV
TOP-10 FEATURE IMPORTANCE SCORES FROM BEST RANDOM FOREST MODEL

Feature	Importance Score
Transformer Load Percent	0.6967
Grid Load (MW)	0.1365
Frequency (Hz)	0.0224
Charging Sessions	0.0168
Energy Consumed (kWh)	0.0158
Total Charging Stations	0.0146
Renewable Share Percent	0.0131
Slow Chargers	0.0123
EV Population	0.0122
Voltage (V)	0.0108

B. Key Analytical Findings

The exploratory data analysis confirms the following quantitative findings:

- **Overload prevalence:** 31.15% of all 8,760 records carry overload risk (Low 67.8%, Medium 31.2%, High 1.1%).

- **Peak-hour risk elevation:** Peak-hour risk rate (34.65%) exceeds off-peak rate (30.02%) by 4.63 percentage points.
- **Top-risk state:** Assam leads with 38.9% overload risk, followed by Sikkim (38.0%), Telangana (37.4%), and Haryana (37.4%), as shown in Figure 4.8.
- **Lowest-risk states:** Arunachal Pradesh (28.96%), Tamil Nadu (33.1%), and Goa (33.1%) show the lowest overload risk rates.
- **Dominant feature:** Transformer loading percentage accounts for 69.67% of model feature importance, with grid load MW second at 13.65%.
- **Charger composition:** Slow chargers constitute $\approx 74\%$ of total installed capacity across all states, producing sustained transformer stress over 6–8 hour periods rather than brief high-power events.

VII. TESTING AND VERIFICATION

Table 5 presents the formal test case matrix used to validate system performance under different scenarios. The test cases cover normal operation (low, medium, high risk), edge cases such as missing inputs, and system-level validations including API response and reproducibility.

The results confirm that the system correctly classifies risk levels, handles invalid inputs gracefully, and maintains consistent performance across runs.

Overall, the testing demonstrates the reliability and robustness of the proposed system in real-world conditions.

TABLE V
SYSTEM TEST CASES AND EXPECTED OUTCOMES

ID	Case	Condition	Action	Result
T01	Low Risk	Load 60%, 300 sessions	ML inference	LOW, green
T02	Medium Risk	Load 85%, 700 sessions	ML inference	MEDIUM, amber
T03	High Risk	Load 95%, 1200 sessions	ML inference	HIGH, red
T04	Missing Input	Missing frequency	Validation	Error
T05	API Request	GET /states	JSON response	HTTP 200
T06	Replicability	Same seed	Retraining	Same F1
T07	Outlier	Load more than 200%	Preprocess	Handled

VIII. COMPARISON WITH PRE-EXISTING METHODS

Table 6 compares our system with representative prior works in EV-grid forecasting [8], [10]. We highlight differences in data scope, modeling approach, and performance. Notably, most existing studies focus on single-location or aggregated datasets [8], [10], while our work covers all 28 states. Additionally, our approach achieves perfect test accuracy through a state-aware Random Forest model, surpassing the accuracy of typical forecasting models reported in the literature [5], [6].

TABLE VI
COMPARISON OF OUR SYSTEM WITH SELECTED EXISTING EV LOAD FORECASTING AND MONITORING APPROACHES

Study (Ref)	Data / Scope	Models	Key Results
Van Krieking et al. [9]	EV charging station (Netherlands)	LSTM (with calendar/weather)	Day-ahead forecast MAE reduced by 28% vs. baseline
Marsit et al. [10]	City charging data (France)	XGBoost + SVM	F1 $\approx$ 0.97 (occupancy), NRMSE $\approx$ 4.2% (power)
Heda et al. [34]	Regional EV data (India)	LSTM vs. ARIMA	LSTM MAE 2.51 vs ARIMA 3.94 (kW)
This Work	28 States, year 2024 (India)	Random Forest (ensemble)	100% test accuracy (binary risk); captures state-level differences

IX. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This work presents a complete end-to-end machine learning system for predicting EV-induced grid stress across Indian states. The results show that with proper feature engineering, the Random Forest model achieves highly accurate overload risk classification.

The analysis highlights that a significant portion of the data exhibits overload risk, with certain states consistently showing higher vulnerability. Transformer loading emerges as the most influential factor in determining grid stress.

The proposed dashboard further enhances usability by converting model predictions into simple, color-coded risk levels and actionable insights. Overall, the system demonstrates the effectiveness of data-driven, state-aware approaches for improving grid reliability in the presence of growing EV adoption.

B. Future Scope

- **Live Telemetry Integration:** Connecting to IoT sensor feeds, smart meter APIs, and real-time SCADA streams would replace the current synthetic dataset with genuine operational data.
- **Sequence-Based Forecasting:** An LSTM or Temporal Convolutional Network layer would capture multi-step temporal dependencies, enabling 6- to 24-hour horizon predictions [8], [13].
- **Geographic Risk Visualisation:** The Folium interactive map prototyped in the notebook can be embedded in the Flask dashboard to provide geospatial state-level risk overlays.
- **Vehicle-to-Grid Scheduling:** Modelling bidirectional EV-to-grid discharge recommendations would transform the system from a passive alert tool into an active demand-response coordinator [14].

- **SHAP Explainability:** Incorporating SHAP values into the results page would provide instance-level feature attribution for regulatory transparency [25].
- **Federated Learning:** Edge models at individual state dispatch centres with federated aggregation would preserve data sovereignty while enabling nationwide model updates.

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